Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB MID TERM

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QUESTION:

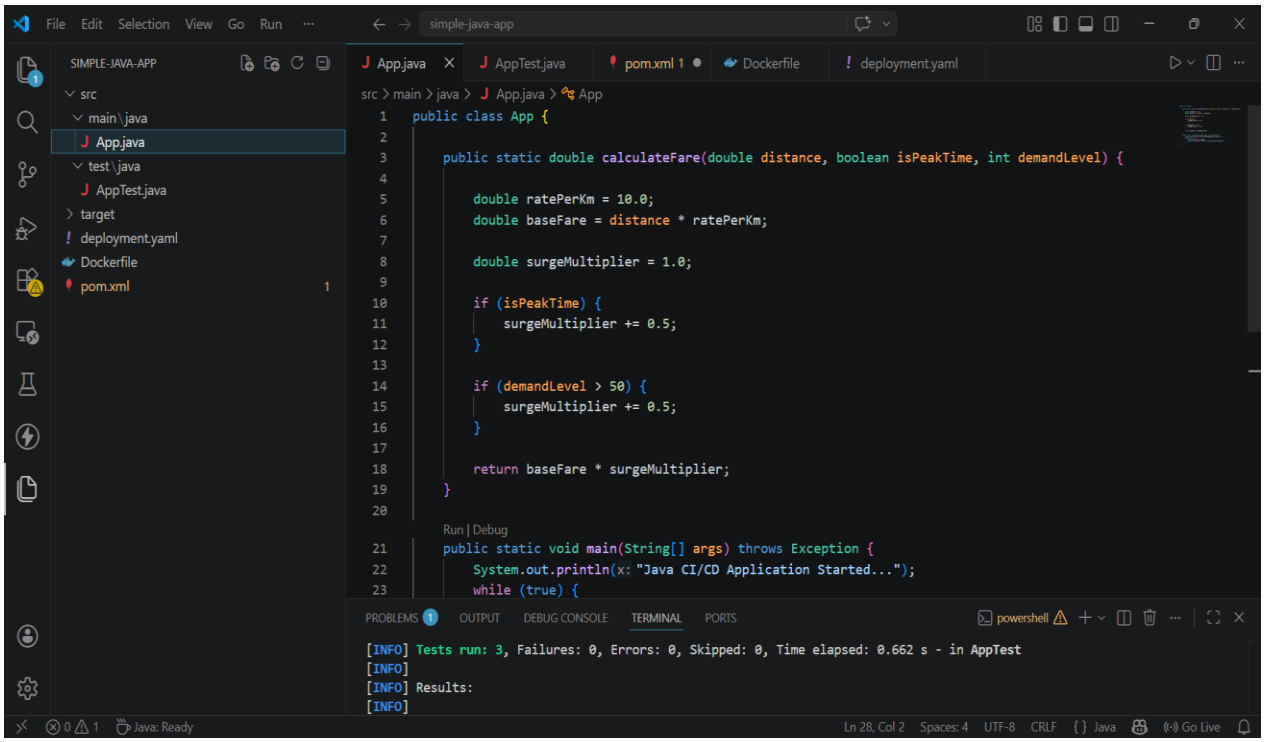
48. Dynamic Pricing System for Ridesharing

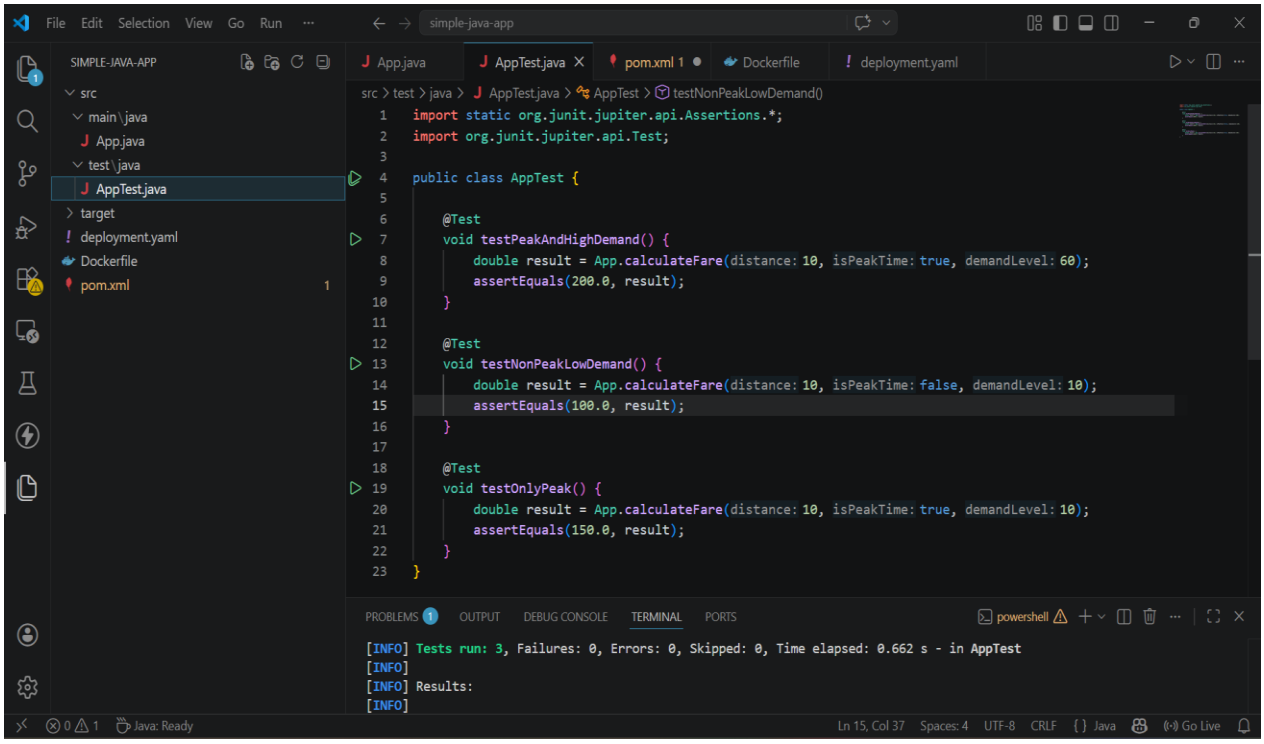
Problem Statement:
Ensure fare prices dynamically adjust based on demand, distance, and time accurately before release.

Tasks:
- **Develop surge pricing algorithm**
- **Write JUnit tests for Peak vs non-peak pricing**
- **Implement CI/CD pipeline with automated testing**
- **Deploy microservice on Docker and Kubernetes**

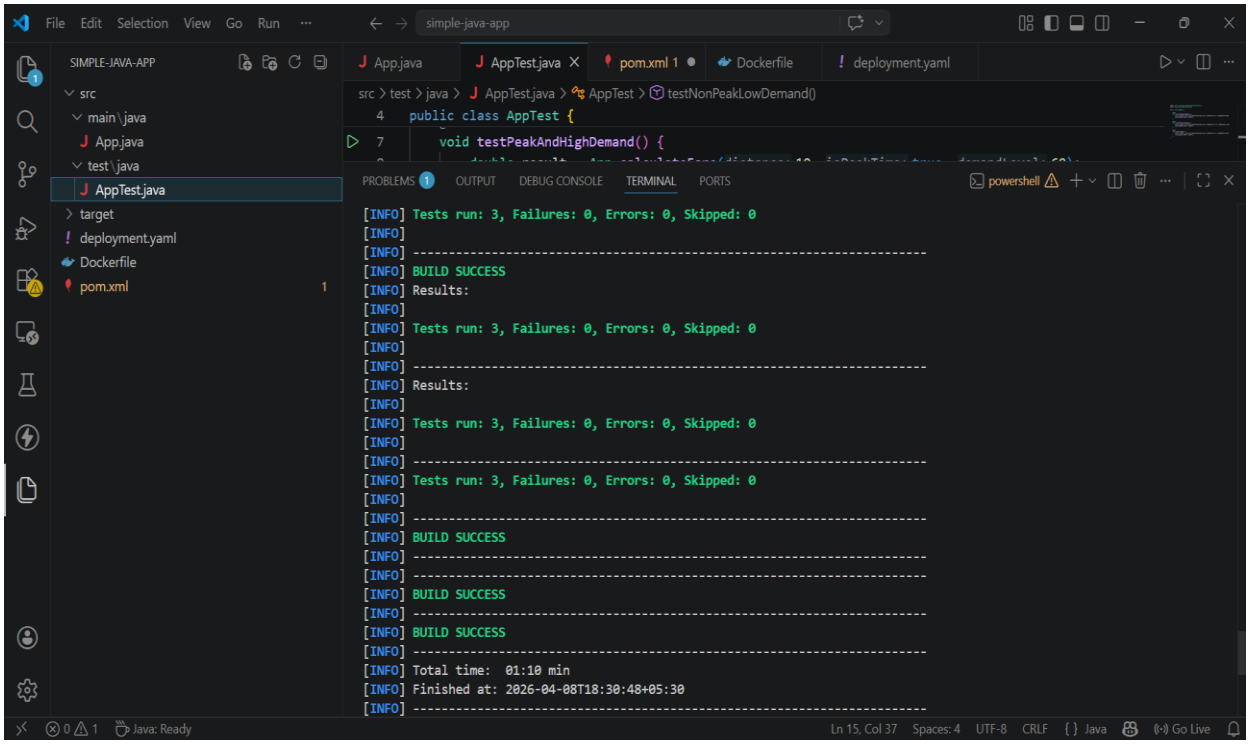
STEPS:

1. JAVA PROGRAM :

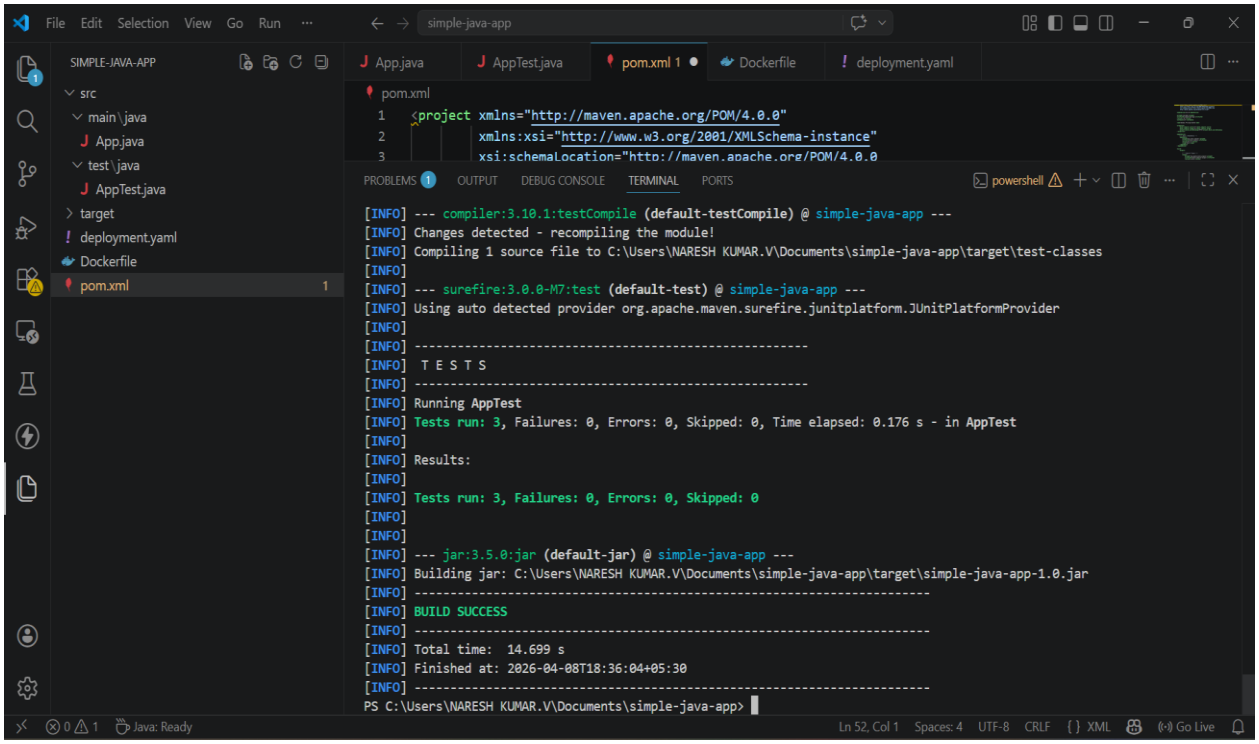




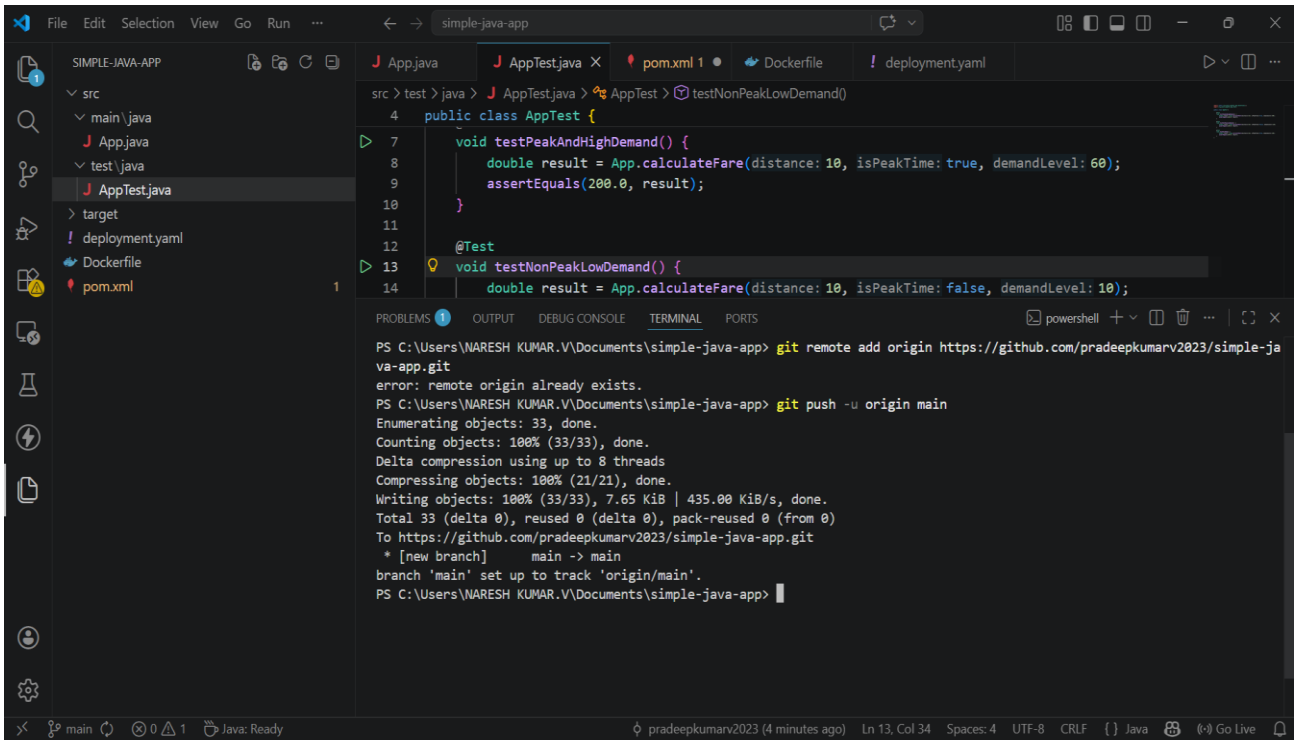
2. JUNIT TESTING(mvn clean test):

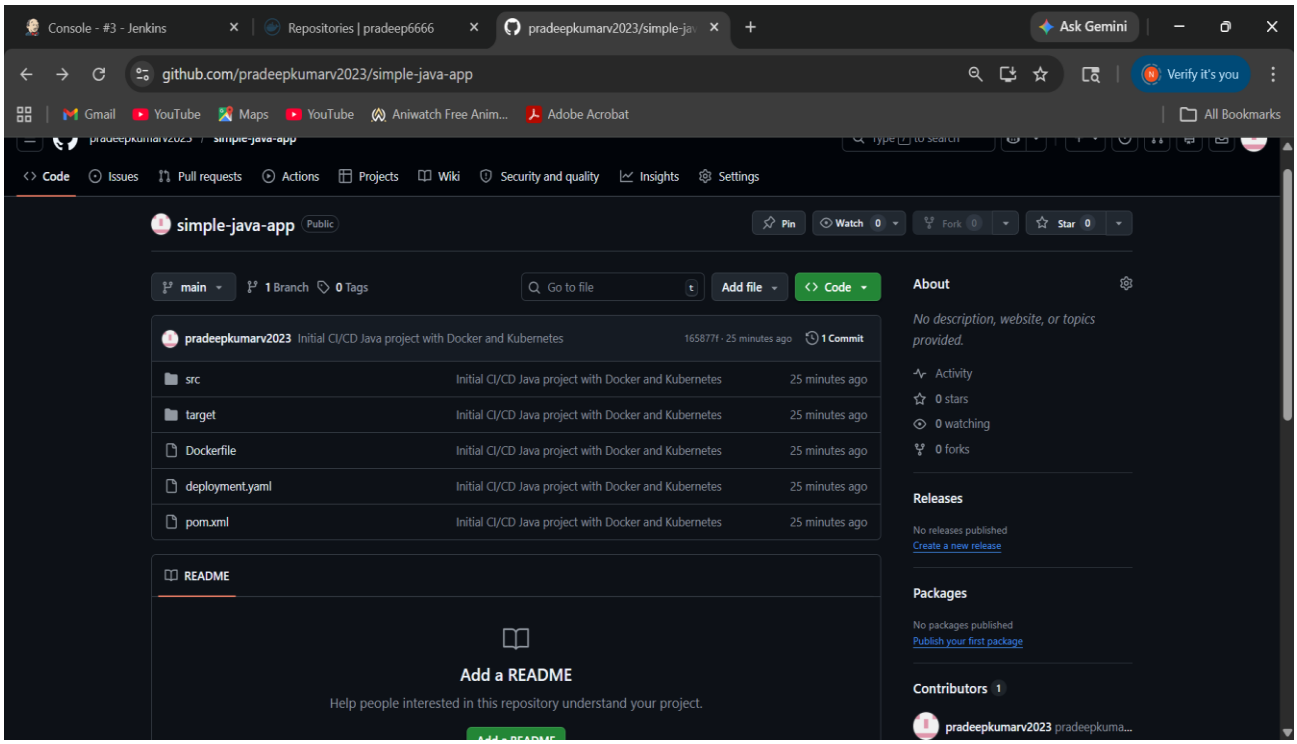


3. MVN clean package:

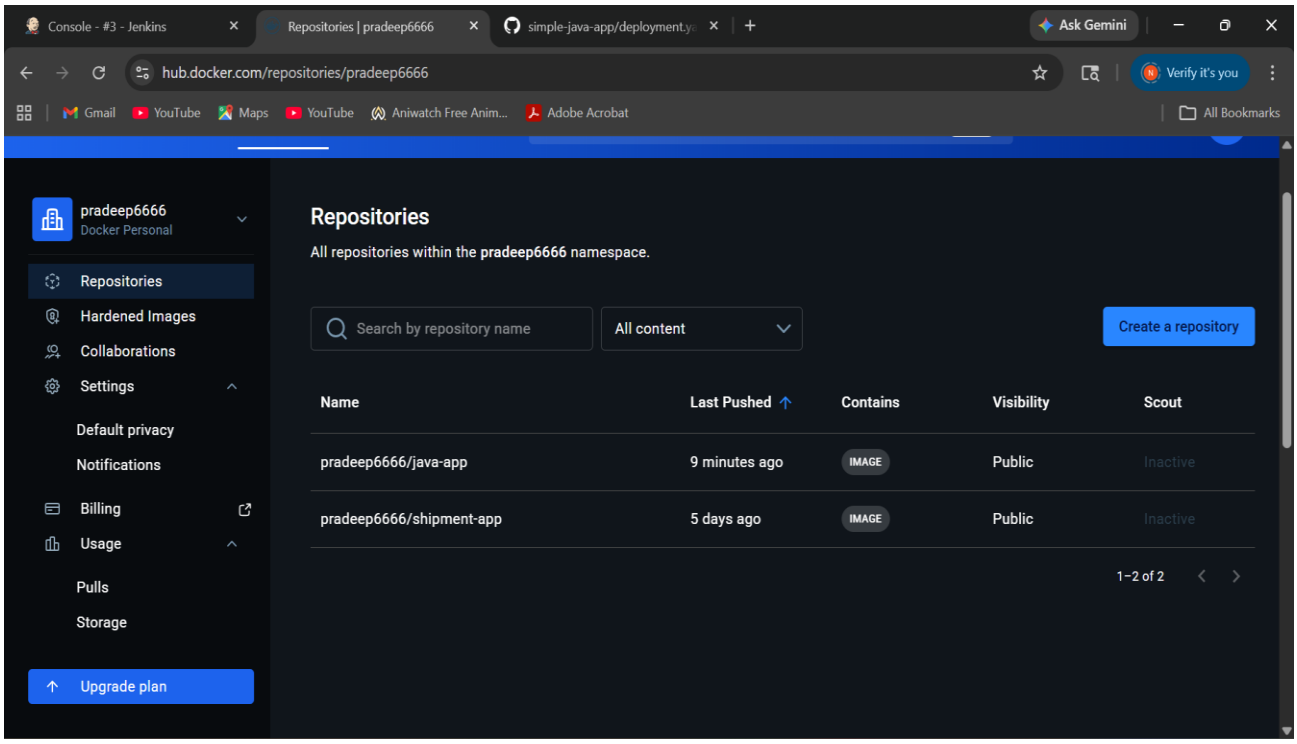


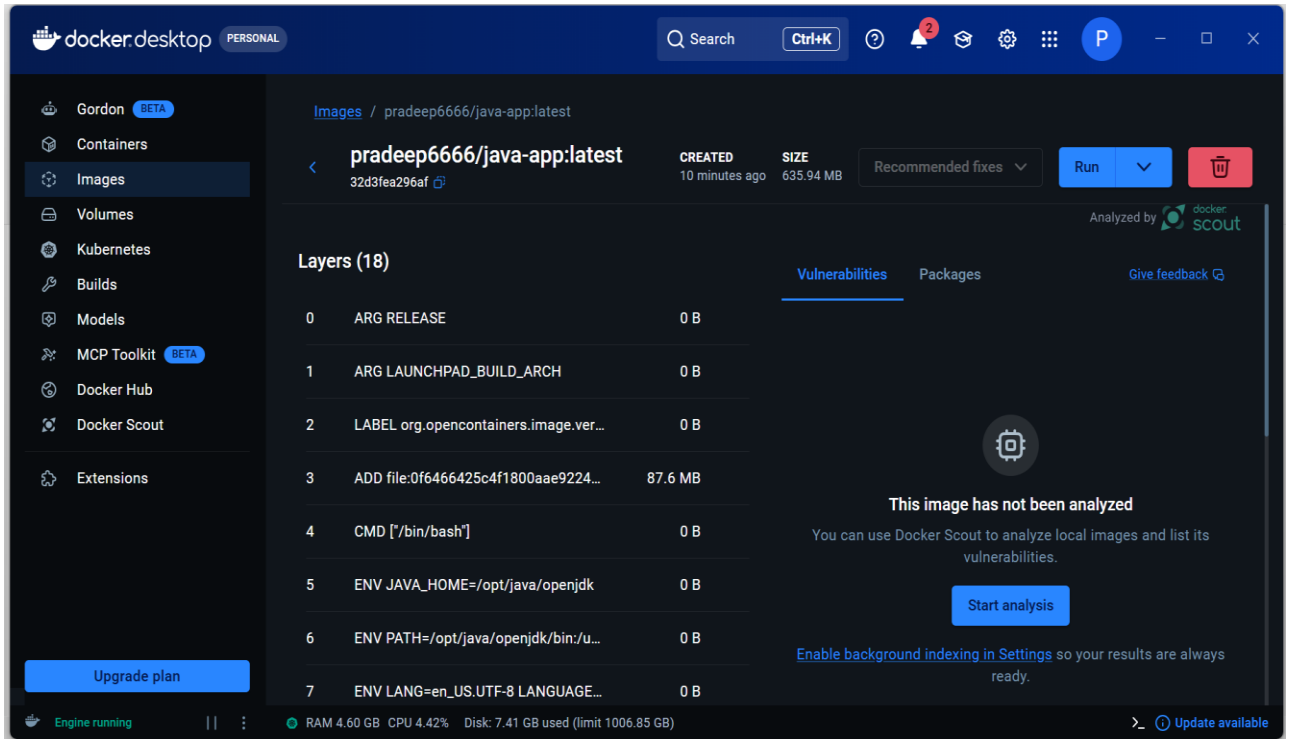
4. GIT PUSH PART:

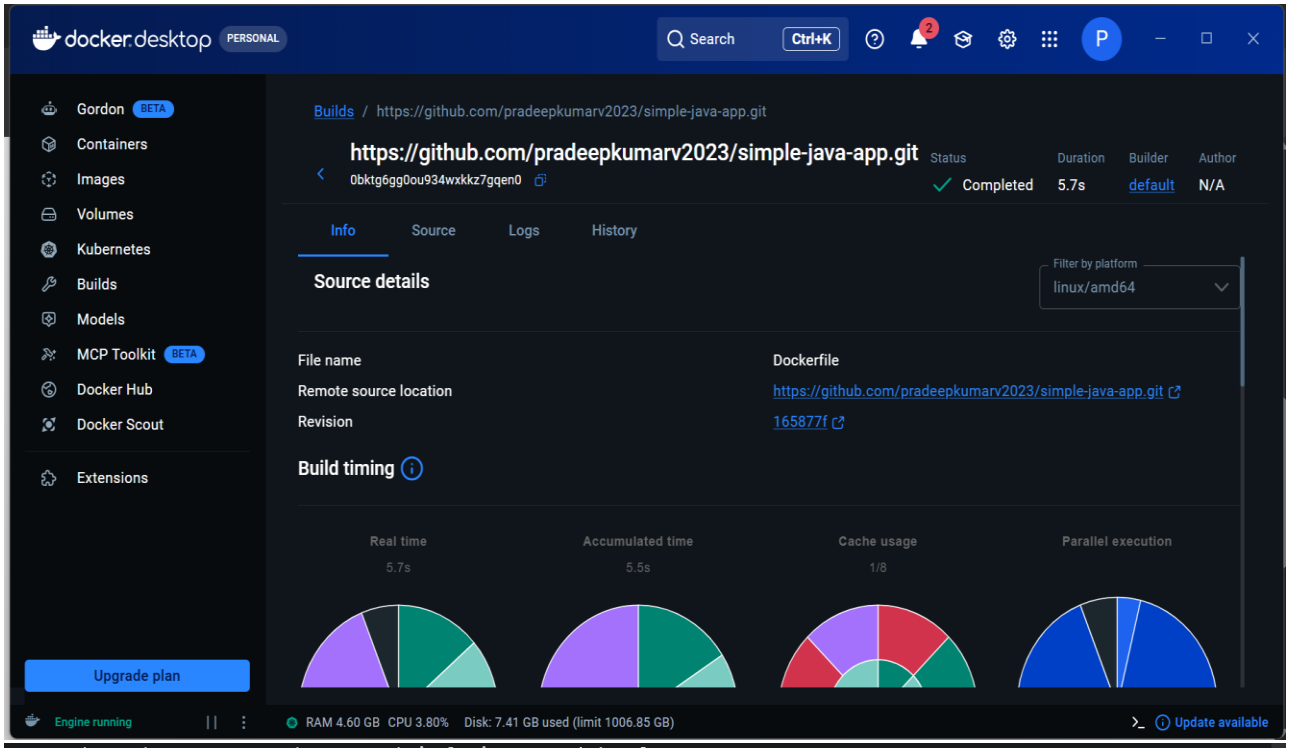




5. DOCKER AND KUBERNETES:







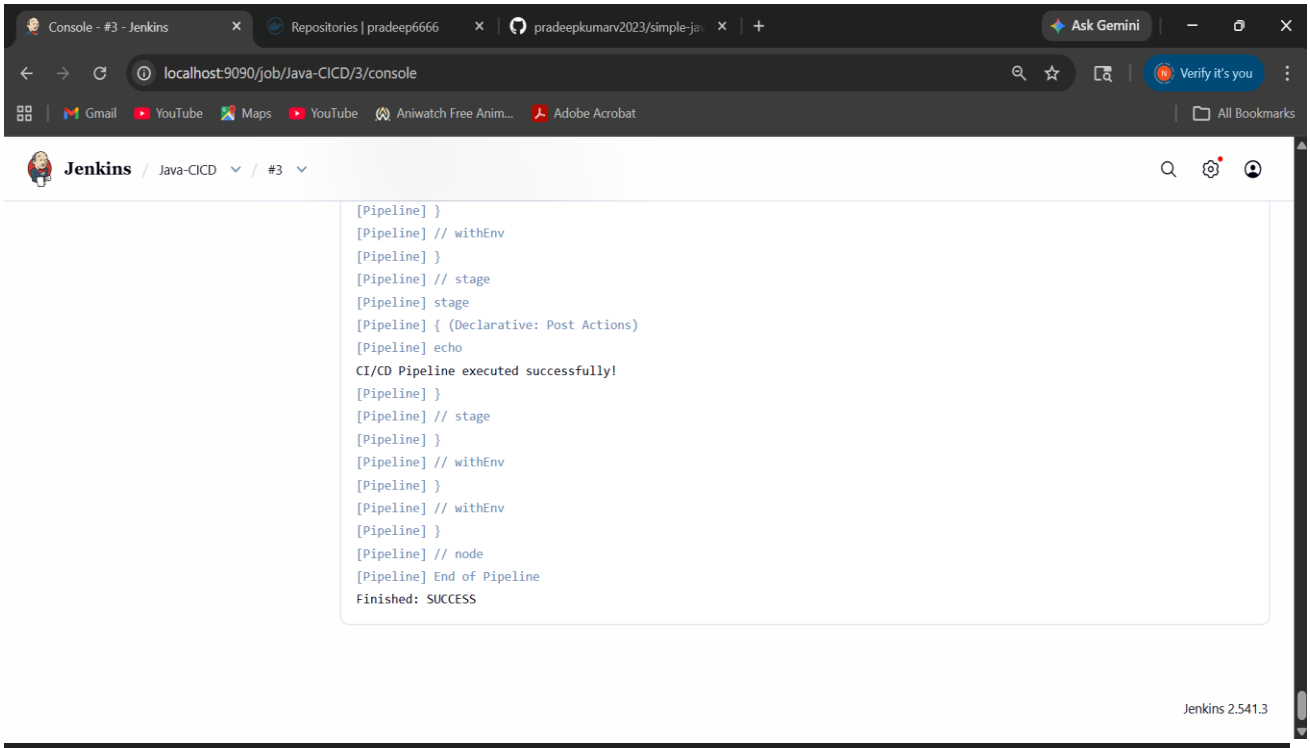

```
PS C:\Users\NARESH KUMAR.V\Documents\simple-java-app> kubectl get svc
NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-service      NodePort    10.96.121.204   <none>        80:30009/TCP   114m
kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP        7d5h
shipment-service  NodePort    10.96.202.199   <none>        80:30008/TCP   5d11h
PS C:\Users\NARESH KUMAR.V\Documents\simple-java-app> kubectl get deployments
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
java-app       0/2     2            0           118m
shipment-app   0/2     2            0           5d11h
```

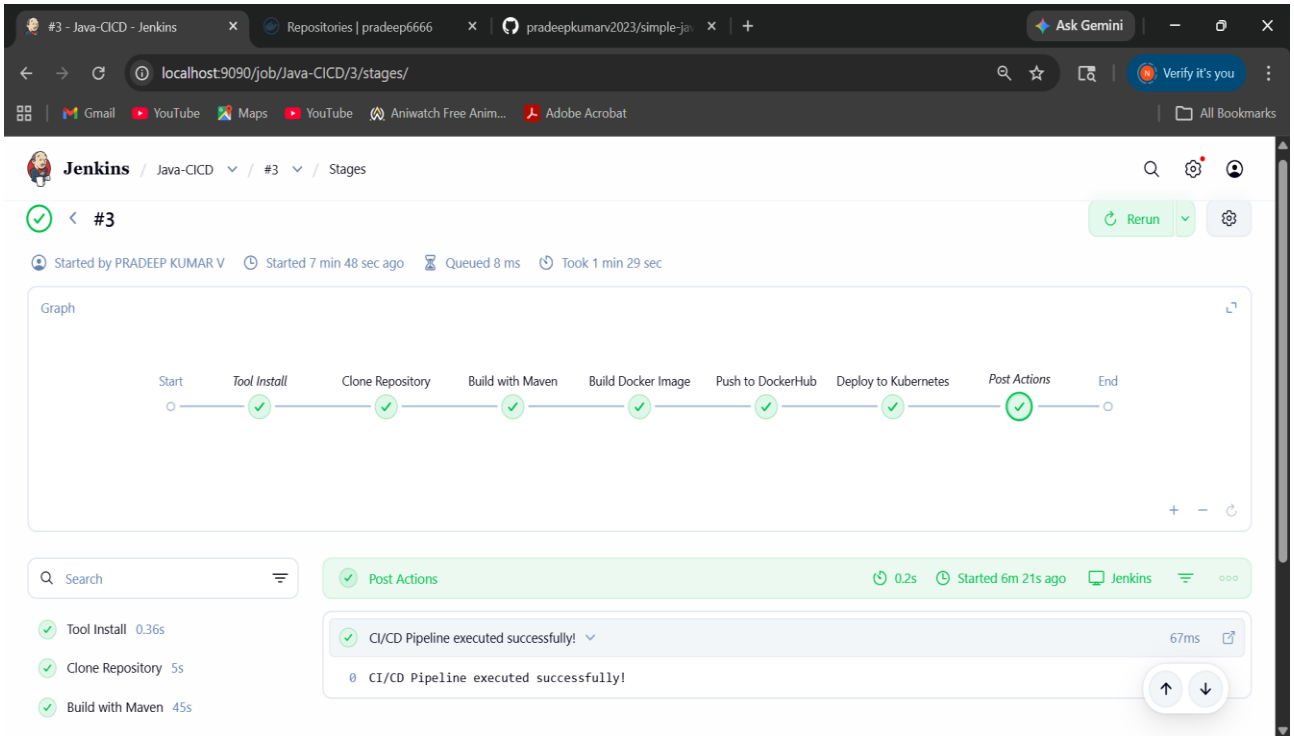
6. **JENKINS:**

PIPELINE CODE:

```
pipeline {
  agent any

  tools {
    maven 'maven-3'
  }

  environment {
    DOCKER_IMAGE = "pradeep6666/java-app:latest"
    KUBECONFIG = "C:\\Users\\NARESH KUMAR.V\\.kube\\config"
  }

  stages {

    stage('Clone Repository') {
      steps {
        git branch: 'main',
        url: 'https://github.com/pradeepkumarv2023/simple-java-app.git'
      }
    }

    stage('Build with Maven') {
      steps {
```

```
            bat 'mvn clean package'
          }
        }

        stage('Build Docker Image') {
          steps {
            bat "docker build -t %DOCKER_IMAGE% ."
          }
        }

        stage('Push to DockerHub') {
          steps {
            withCredentials([usernamePassword(
              credentialsId: 'dockerhub-creds',
              usernameVariable: 'DOCKER_USER',
              passwordVariable: 'DOCKER_PASS'
            )]) {
              bat """
              docker login -u %DOCKER_USER% -p %DOCKER_PASS%
              docker push %DOCKER_IMAGE%
              """
            }
          }
        }

        stage('Deploy to Kubernetes') {
          steps {
            bat '"C:\\Program Files\\Docker\\Docker\\resources\\bin\\kubectl.exe" apply -f
deployment.yaml'
          }
        }
      }

      post {
        success {
          echo 'CI/CD Pipeline executed successfully!'
        }
        failure {
          echo 'Pipeline failed. Please check logs.'
        }
      }
    }
```


Console - #3 - Jenkins
Repositories | pradeep6666
pradeepkumarv2023/simple-ja
Ask Gemini
localhost:9090/job/Java-CICD/3/console
Verify it's you
Gmail
YouTube
Maps
YouTube
Aniwatch Free Anim...
Adobe Acrobat
All Bookmarks
Jenkins
Java-CICD
#3
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.3


#3 - Java-CICD - Jenkins
Repositories | pradeep6666
pradeepkumarv2023/simple-ja
Ask Gemini
localhost:9090/job/Java-CICD/3/stages/
Verify it's you
Gmail
YouTube
Maps
YouTube
Aniwatch Free Anim...
Adobe Acrobat
All Bookmarks
Jenkins
Java-CICD
#3
Stages
#3
Rerun
Started by PRADEEP KUMAR V
Started 7 min 48 sec ago
Queued 8 ms
Took 1 min 29 sec
Graph
Start
Tool Install
Clone Repository
Build with Maven
Build Docker Image
Push to DockerHub
Deploy to Kubernetes
Post Actions
End
Search
Tool Install 0.36s
Clone Repository 5s
Build with Maven 45s
Post Actions
0.2s
Started 6m 21s ago
Jenkins
CI/CD Pipeline executed successfully!
67ms
0 CI/CD Pipeline executed successfully!